Solve

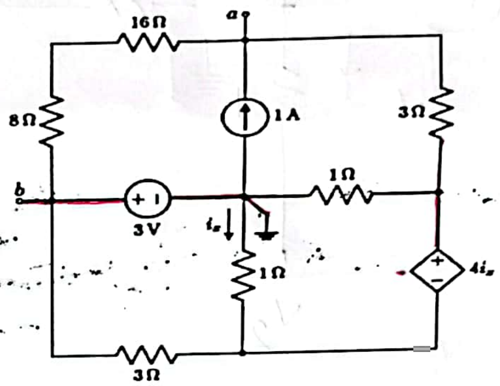
ID:	Name:

Brac University
Semester: Spring 2025
Course Code: CSE250
Circuits and Electronics
Section: 07

Assessment: Assignment 3
Full Marks: 20

Question 1 of 2

Consider the following circuit with terminal a and b. Currently no load is connected to the terminals.

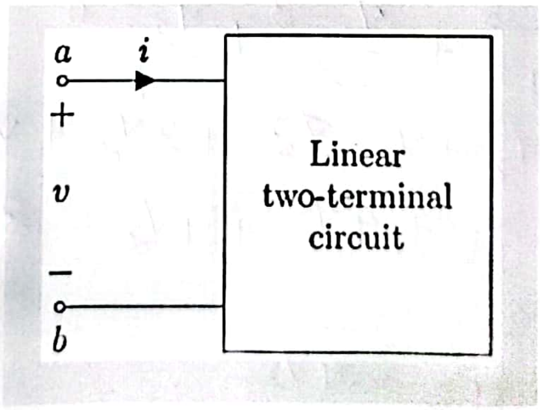


a) Draw the equivalent circuit using Thevenin theorem. ~~[3]~~ 6

b) Find the Norton equivalent current. ~~[2]~~ 4

Question 2 of 2:

Following is the unknown circuit. Except the two terminal other part of the circuit is not accessible. After applying 30 V to the terminal, 1 A current was measured with terminal a-b. Similarly, after applying 50 V, 3 A current was measured.

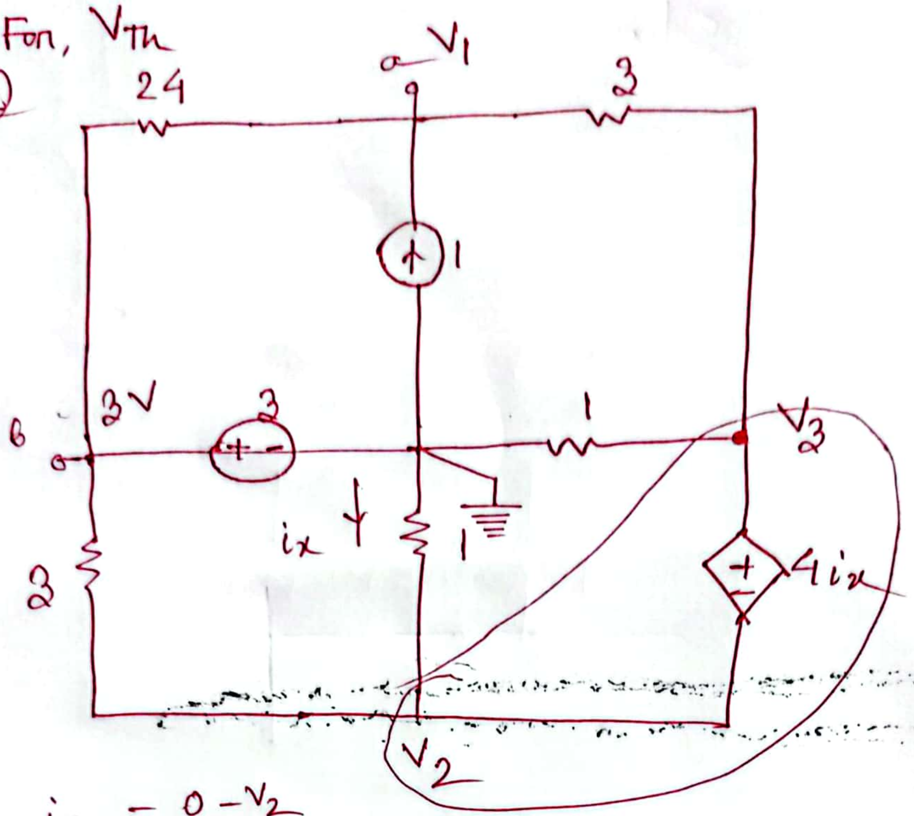


a. Find the I-V equation of the unknown circuit. ~~[3]~~ 6

b. From the equation derive the equivalent circuit (Either in Voltage Source- Resistance form or Current Source-Resistance form). ~~[2]~~ 4

1. For, V_{Th}

(a)



$$i_x = \frac{0 - V_2}{1}$$

$$i_x = -V_2$$

At node, ①,
$$\frac{V_1 - 3}{24} + \frac{V_1 - V_3}{3} - 1 = 0$$

$$V_1 - 3 + 8V_1 - 8V_3 - 24 = 0$$

$$\Rightarrow 9V_1 - 8V_3 = 27 \longrightarrow (\mathrm{I})$$

at supernode, ② & ③,

$$\frac{V_2 - 0}{1} + \frac{V_2 - 3}{3} + \frac{V_3 - 0}{1} + \frac{V_3 - V_1}{3} = 0$$

$$\Rightarrow 3V_2 + V_2 - 3 + 3V_3 + V_3 - V_1 = 0$$

$$\Rightarrow -V_1 + 4V_2 + 4V_3 = 3 \longrightarrow (\mathrm{II})$$

$$V_3 - V_2 = 4i_x$$

$$\Rightarrow V_3 - V_2 = -4V_2$$

$$\Rightarrow V_3 + 3V_2 = 0$$

$$\Rightarrow 3V_2 + V_3 = 0 \longrightarrow (\mathrm{III})$$

$V_1 = 6\,V$

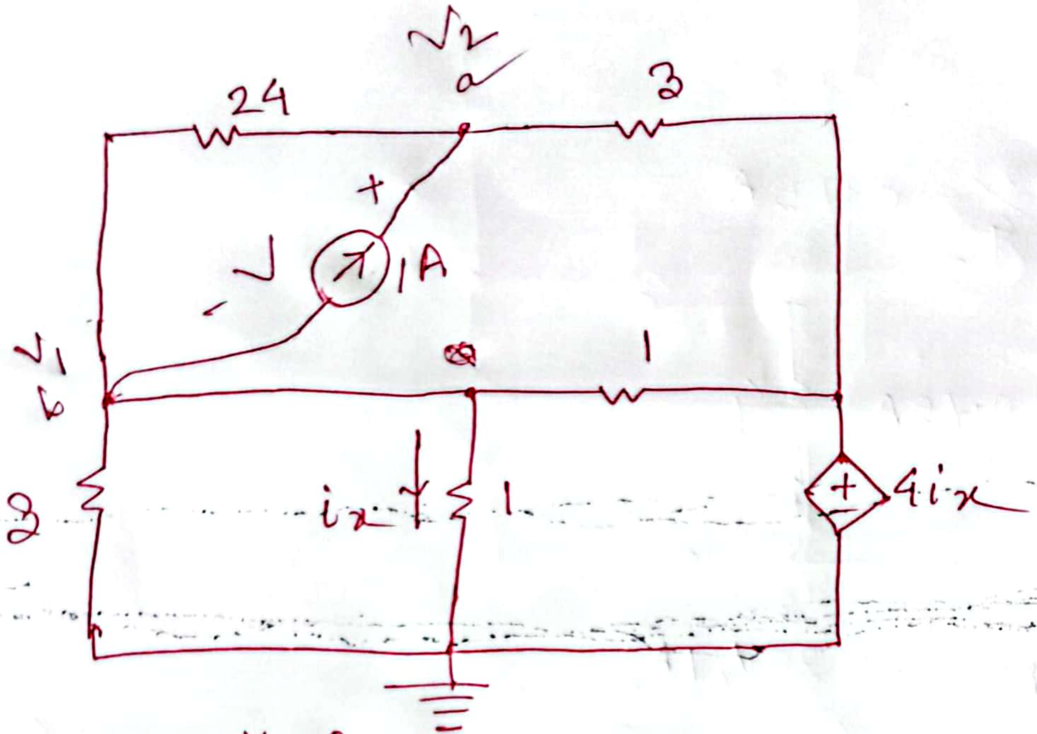
$V_2 = -1.125\,V$

$V_3 = 3.375\,V$

$$V_{Th} = V_{ab} = V_a - V_b$$

$$= 6 - 3 = 3V$$

For R_{Th}



$$i_x = \frac{V_1 - 0}{1} = V_1$$

at node, ①,

$$\frac{V_1}{3} + 1 + \frac{V_1 - V_2}{24} + \frac{V_1}{1} + \frac{V_1 - 4i_x}{1} = 0$$

$$\Rightarrow \frac{V_1}{3} + 1 + \frac{V_1 - V_2}{24} + V_1 - 3V_1 = 0$$

$$\Rightarrow 8V_1 + 24 + V_1 - V_2 - 48V_1 = 0$$

$$\Rightarrow -39V_1 - V_2 = -24 \longrightarrow (i)$$

at node ②,

$$-1 + \frac{V_2 - V_1}{24} + \frac{V_2 - 4i_x}{3} = 0$$

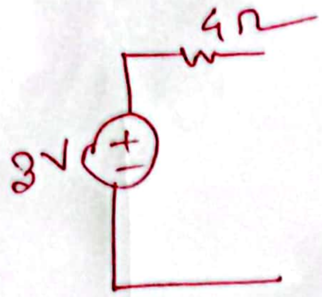
$$\Rightarrow -1 + \frac{V_2 - V_1}{24} + \frac{V_2 - 4V_1}{3} = 0$$

$$\Rightarrow -24 + V_2 - V_1 + 8V_2 - 32V_1 = 0$$

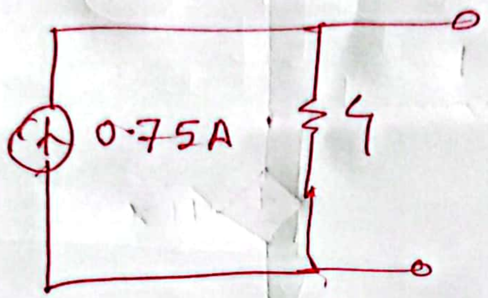
$$\Rightarrow -33V_1 + 9V_2 = 24 \longrightarrow (ii)$$

$V_1 = 0.5$, $V_2 = 4.5V$

$V = V_2 - V_1 = 4V$, $\quad R_{Th} = \frac{V}{1} = \frac{4}{1}\Omega = 4\Omega$



b) $I_N = \frac{3}{4} = 0.75\,A$



2. (a) First point : (30V, 1A)
Second " : (50V, 3A)

$y \rightarrow I \quad x \rightarrow V$

$$\frac{I-1}{1-3} = \frac{V-30}{30-50} \left[\frac{y-y_1}{y_1-y_2} = \frac{x-x_1}{x_1-x_2}\right]$$

$$\Rightarrow \frac{I-1}{\cancel{-2}} = \frac{V-30}{\cancel{-20}_{10}}$$

$$\Rightarrow I - 1 = \frac{V}{10} - 3$$

$$\Rightarrow I = \frac{V}{10} - 2$$

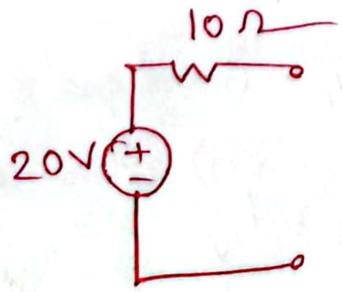
(b)

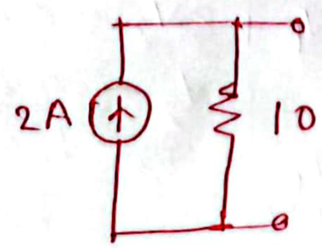
$$I = \frac{V}{R} - \frac{V_s}{R}$$

$$\frac{1}{R} = \frac{1}{10} \qquad \frac{V_s}{R} = 2$$

$$R = 10\,\Omega, \qquad \frac{V_s}{10} = 2$$

$$\Rightarrow V_s = 20\,V$$





$I_s = \frac{V_s}{R} = \frac{20}{10} = 2$